Demo: Homopolar Motor

ChatGPT edition — adult-controlled, five-second AA-cell trials

Magnetic force and torque in three parts

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

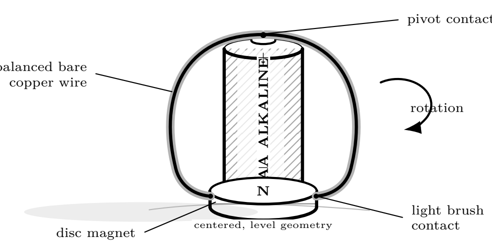
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- One fresh AA alkaline cell
- One neodymium disc magnet
- Bare copper wire
- Pliers
- Timer

Predict first

If the magnet flips, should rotation reverse? What if both cell and magnet flip?



Investigate

1. Adult counts and mounts the magnet on the negative cell end. Bend balanced wire to touch the positive button and barely brush the magnet edge.
2. Clear the area; adult makes contact for no more than five seconds while learner observes direction.
3. Disconnect, check wire and cell temperature, and allow cooling.
4. Flip the magnet and repeat once. Store the magnet securely.

Explain from the evidence

Current crossing the magnet's field experiences force; the geometry turns that force into torque. Reversing either current or field reverses rotation; reversing both preserves it. This is nearly a direct short, so most input can become heat. That is why the trial is brief, adult-controlled, AA alkaline only, and never lithium.

Change one thing

Predict four combinations of cell direction and magnet direction before testing only the minimum needed to check the pattern.

Evidence record	
Prediction	
One change	
Observation	
Claim + because	

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.